CMPG 322 - Decision Support Systems (DSS) 2
SU6 - Waiting Lines and Queuing Theory



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Outline

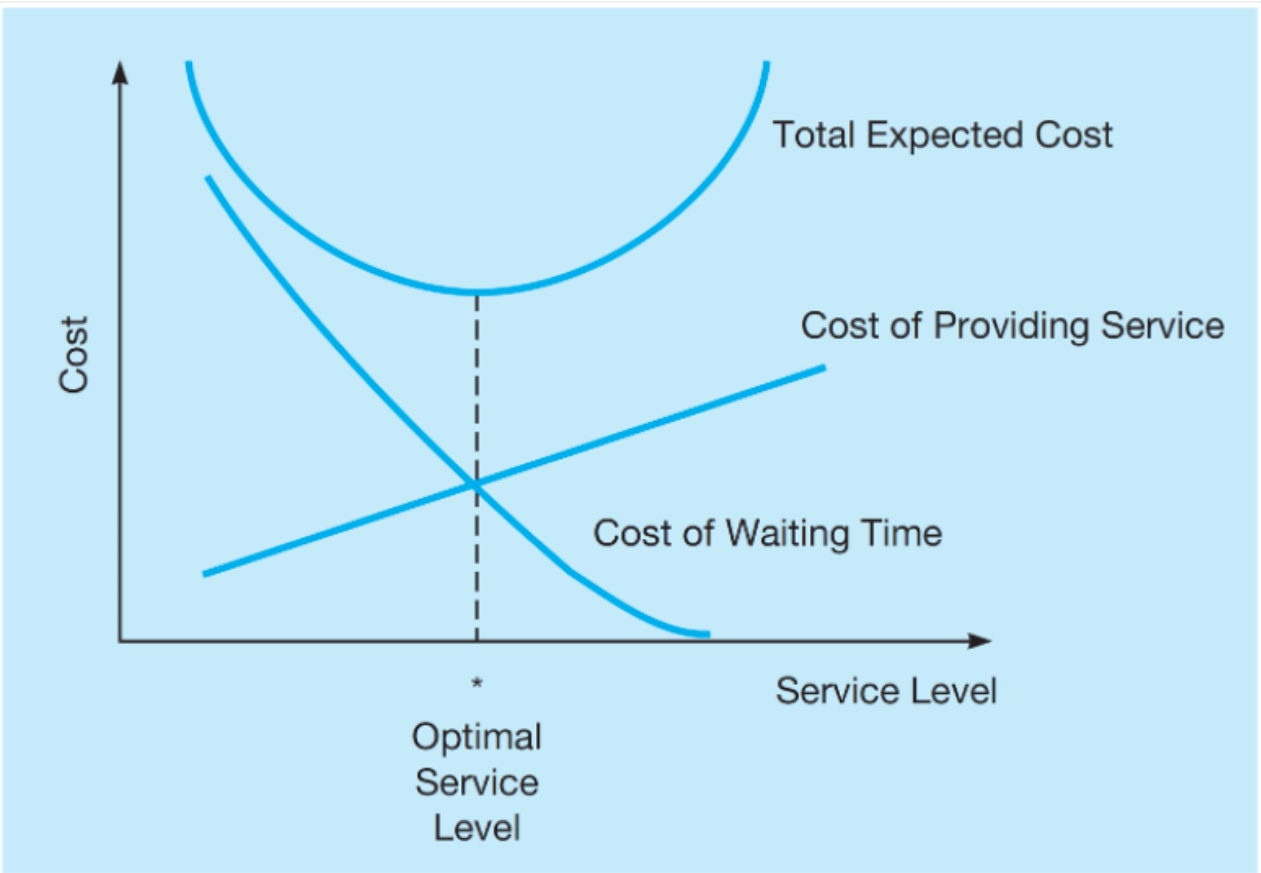
- The trade-off curves for cost-of-waiting time and cost of service.
- The basic queuing system configurations and the three parts of a queuing system;
- Waiting lines for single-channel models with exponential service times and infinite calling populations;
- Waiting lines for multichannel models with exponential service times and infinite calling populations;
- Waiting lines for single-channel models with deterministic service times and infinite calling populations;
- Waiting lines for single-channel models with exponential service times and finite calling populations; and
- Little's Flow Equations.

- Queuing theory is the study of waiting lines
  - One of the oldest and most widely used quantitative analysis techniques
- The three basic components of a queuing process
  - Arrivals
  - Service facilities
  - The actual waiting line
- Analytical models of waiting lines can help managers evaluate the cost and effectiveness of service systems

Waiting line cost

- Most waiting line problems are focused on finding the ideal level of service a firm should provide
- Often try to find the balance between two extremes
  - A large staff and many service facilities
  - The minimum number of service facilities
- Service facilities are evaluated on their total expected cost = service costs + waiting costs
  - Find the service level that minimizes the total expected cost

FIGURE 12.1 Queuing Costs and Service Levels



Example: Three Rivers Shipping Company

- Three Rivers Shipping operates a docking facility on the Ohio River
  - An average of 5 ships arrive to unload their cargos each shift
  - Idle ships are expensive
  - More staff can be hired to unload the ships, but that is expensive as well
- Three Rivers Shipping Company wants to determine the optimal number of teams of stevedores to employ each shift to obtain the minimum total expected cost

Example: Three Rivers Shipping Company

Number of Teams of Stevedores Working	**1**	**2**	**3**	**4**
(a) Avg. number of ships arriving per shift	5	5	5	5
(b) Avg. time each ship waits to be unloaded (hours)	7	4	3	2
(c) Total ship hours lost per shift (a × b)	35	20	15	10
(d) Estimated cost per hour of idle ship time	\$1,000	\$1,000	\$1,000	\$1,000
(e) Value of ship's lost time or waiting cost (c × d)	\$35,000	\$20,000	\$15,000	\$10,000
(f) Stevedore team salary or service cost	\$6,000	\$12,000	\$18,000	\$24,000
(g) Total expected cost (e + f)	\$41,000	\$32,000	\$33,000	\$34,000

Table: Three Rivers Shipping Company Waiting Line Cost Analysis

- There are three parts to a queuing system
  1. Arrivals or inputs to the system (sometimes referred to as the calling population)
  2. The queue or waiting line itself
  3. The service facility

- Size of calling population: unlimited or limited
- Pattern of arrivals: known pattern or random (Poisson distribution)
- Behaviour:
  - Patient
  - Balking
  - Reneging

- Limited or unlimited
- Queue discipline refers to the rule by which customers in the line receive service
  - FIFO
  - Other rules can be applied to select which customers enter which queue, but may apply FIFO once they are in the queue

- Configuration: single- or multichannel system
- Pattern of service times: single- or multiphase system

FIGURE 12.2 Four Basic Queuing System Configurations

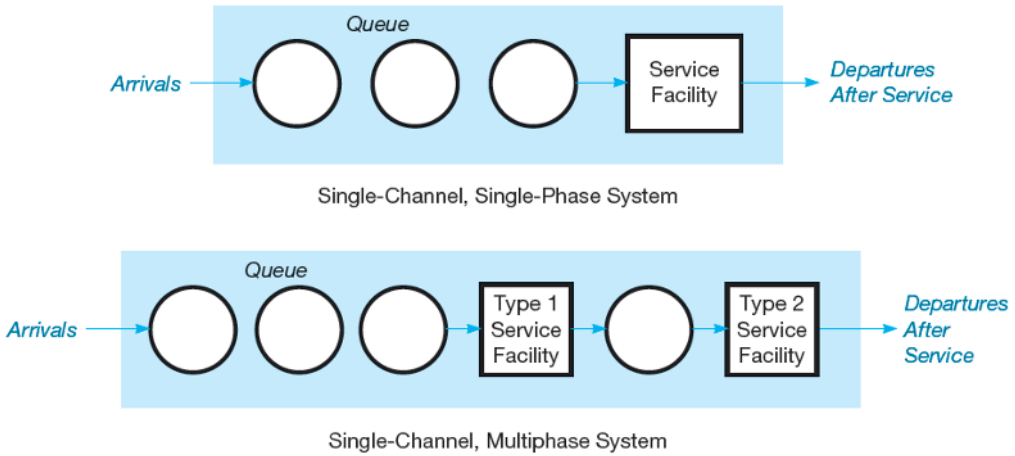
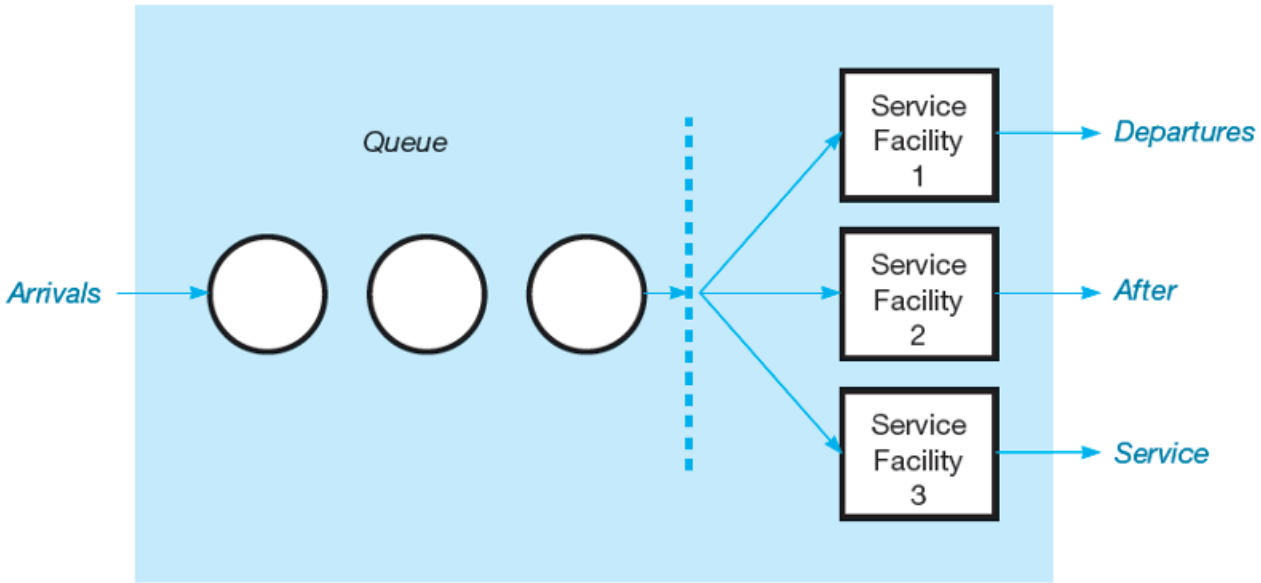
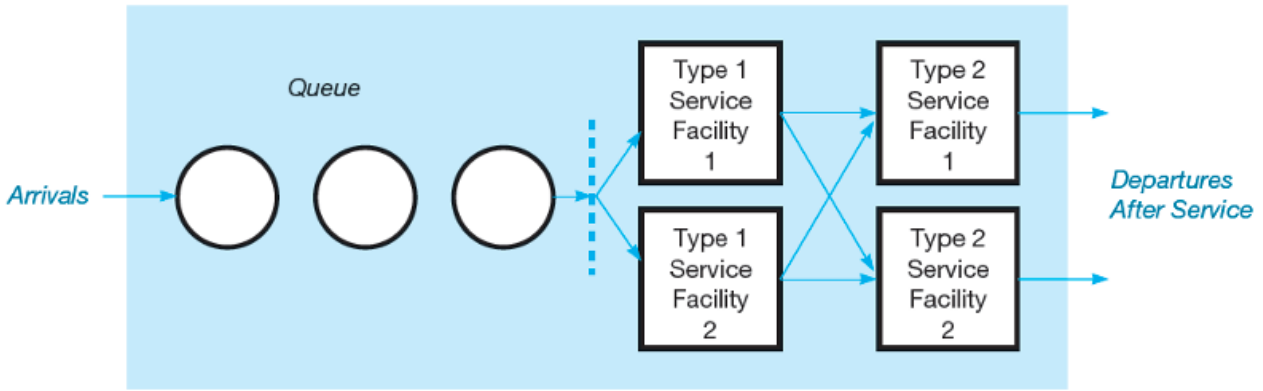


FIGURE 12.2 Four Basic Queuing System Configurations



Multichannel, Single-Phase System

FIGURE 12.2 Four Basic Queuing System Configurations



Service time distribution:

- Service patterns either constant or random
- Constant service times often machine controlled
- Generally service times are randomly distributed according to a negative exponential probability distribution

Identifying models using Kendall notation

- A notation for queuing models that specifies the pattern of arrival, the service time distribution, and the number of channels
- Basic three-symbol **Kendall notation**:

Arrival distribution / Service time distribution / Number of service channels

- Specific letters used to represent probability distributions:
  - M = Poisson distribution for number of occurrences
  - D = constant (deterministic) rate
  - G = general distribution with mean and variance known

Identifying models using Kendall notation

- A single-channel model with Poisson arrivals and exponential service times would be represented by:

$$M/M/1$$

- When a second channel is added:

$$M/M/2$$

- An m-channel system with Poisson arrivals and constant service time would be:

$$M/D/m$$

- A four-channel system with Poisson arrivals and normally distributed service times would be:

$$M/G/4$$

Assumptions of the model:

1. **Arrivals are served on a FIFO basis**
2. **There is no balking or reneging**
3. **Arrivals are independent of each other but the arrival rate is constant over time**
4. **Arrivals follow a Poisson distribution**
5. **Service times are variable and independent but the average is known**
6. **Service times follow a negative exponential distribution**
7. **Average service rate is greater than the average arrival rate**

When these assumptions are met, we can develop a series of equations that define the queue's **operating characteristics**

Queuing Equations

- **Let**

  $\lambda =$ mean number of arrivals per time period

  $\mu =$ mean number of customers or units served per time period

- **The same time period must be used for the arrival rate and service rate**

1. The average number of customers or units in the system, L

$$L = \frac{\lambda}{\mu - \lambda}$$

2. The average time a customer spends in the system, W

$$W = \frac{1}{\mu - \lambda}$$

3. The average number of customers in the queue, L_q

$$L_q = \frac{\lambda^2}{\mu(\mu - \lambda)}$$

4. The average time a customer spends waiting in the queue, W_q

$$W_q = \frac{\lambda}{\mu(\mu - \lambda)}$$

5. The utilization factor for the system, ρ, the probability the service facility is being used

$$\rho = \frac{\lambda}{\mu}$$

6. The percent idle time, P_0, or the probability no one is in the system

$$P_0 = 1 - \frac{\lambda}{\mu}$$

7. The probability that the number of customers in the system is greater than k, $P_{n>k}$

$$P_{n>k} = \left(\frac{\lambda}{\mu}\right)^{k+1}$$

M/M/1 example: Arnold's Muffler Shop

- Arnold's mechanic can install mufflers at a rate of 3 per hour
- Customers arrive at a rate of 2 per hour

$$\lambda = 2 \quad \mu = 3$$

- Average number of cars in the system:

$$L = \frac{\lambda}{\mu - \lambda} = \frac{2}{3-2} = \frac{2}{1} = 2 \text{ cars in the system on average}$$

- Average time a car spends in the system:

$$W = \frac{1}{\mu - \lambda} = \frac{1}{3-2} = 1 \text{ hour that an average car spends in the system}$$

M/M/1 example: Arnold's Muffler Shop

- Average number of cars in the queue:

$$L_q = \frac{\lambda^2}{\mu(\mu-\lambda)} = \frac{2^2}{3(3-2)} = \frac{4}{3(1)} = 1.33 \text{ cars waiting in line on average}$$

- Average waiting time in the queue:

$$W_q = \frac{\lambda}{\mu(\mu-\lambda)} = \frac{2}{3(3-2)} = \frac{2}{3} \text{ hour} = 40 \text{ minutes average waiting time per car}$$

- Utilization factor:

$$\rho = \frac{\lambda}{\mu} = \frac{2}{3} = 0.67 = 67\% \text{ of time mechanic is busy}$$

- Probability of 0 cars in the system:

$$P_0 = 1 - \frac{\lambda}{\mu} = 1 - \frac{2}{3} = 0.33$$

Probability of more than k cars in the system

k	$P_{n>k} = (2/3)^{k+1}$
0	0.667
1	0.444
2	0.296
3	0.198
4	0.132
5	0.088
6	0.058
7	0.039

*Note: $P_{n>0} = 1 - P_0 = 1 - 0.33 = 0.667$
This implies that there is a 19.8% chance that more than 3 cars are in the system

Introducing costs into the model

- **The total service cost** is:

$$\text{Total service cost} = mC_s$$

where

m = number of channels

C_s = service cost (labor cost) of each channel

The total waiting cost is

- Based on time in the system:

$$\begin{aligned}\text{Total waiting cost} &= (\text{Total time spent waiting by all arrivals}) \times (\text{Cost of waiting}) \\ &= (\text{Number of arrivals}) \times (\text{Average wait per arrival}) \times C_w \\ &= (\lambda W) C_w\end{aligned}$$

- Based on time in the queue:

$$\text{Total waiting cost} = (\lambda W_q) C_w$$

Total cost of the queuing system

- Based on time in the system:

$$\text{Total cost} = \text{Total service cost} + \text{Total waiting cost} = mC_s + \lambda W C_w$$

- Based on time in the queue:

$$\text{Total cost} = mC_s + \lambda W_q C_w$$

M/M/1 example: Arnold's Muffler Shop

- Arnold estimates the cost of customer waiting time in line is \$50 per hour

  Total daily waiting cost $= (8 \text{ hours/day}) \times \lambda W_q C_w = (8)(2)(\frac{2}{3})(50) = \$533.33/\text{day}$

- The mechanic's wage is \$15 per hour

  Total daily service cost $= (8 \text{ hours/day}) \times mC_s = (8)(1)(15) = \$120/\text{day}$

- Total cost of the system is

  $$\text{Total daily cost} = \$533.33 + \$120 = \$653.33$$

- Arnold is thinking about hiring a different mechanic who can install mufflers at a faster rate
- The new operating characteristics would be:

$$\lambda = 2 \text{ cars/hour} \quad \mu = 4 \text{ cars/hour}$$

- Average number of cars in the system:

$$L = \frac{\lambda}{\mu - \lambda} = \frac{2}{4-2} = \frac{2}{2} = 1 \text{ car in the system on average}$$

- Average time in the system:

$$W = \frac{1}{\mu - \lambda} = \frac{1}{4-2} = 0.5 \text{ hour}$$

M/M/1 example: Arnold's Muffler Shop

- Average number of cars in the queue:

$$L_q = \frac{\lambda^2}{\mu(\mu-\lambda)} = \frac{2^2}{4(4-2)} = \frac{4}{8} = 0.5 \text{ cars waiting in line}$$

- Average waiting time in the queue:

$$W_q = \frac{\lambda}{\mu(\mu-\lambda)} = \frac{2}{4(4-2)} = \frac{2}{8} = 0.25 \text{ hour} = 15 \text{ minutes}$$

- Utilization factor:

$$\rho = \frac{\lambda}{\mu} = \frac{2}{4} = 0.5 = 50\% \text{ of time mechanic is busy}$$

- Probability of 0 cars in the system:

$$P_0 = 1 - \frac{\lambda}{\mu} = 1 - \frac{2}{4} = 0.5$$

Probability of more than k cars in the system

k	$P_{n>k} = (2/4)^{k+1}$
0	0.500
1	0.250
2	0.125
3	0.062
4	0.031
5	0.016
6	0.008
7	0.004

- The customer waiting cost is still \$50 per hour

$$\text{Total daily waiting cost} = (8)(2)(0.25)(50) = \$200.00/\text{day}$$

- The new mechanic is more expensive at \$20 per hour:

$$\text{Total daily service cost} = (8)(1)(20) = \$160/\text{day}$$

- Total cost of the system is:

$$\text{Total daily cost} = \$200 + \$160 = \$360$$

- The total time spent for 16 clients per day was previously:

$$(16 \text{ cars/day}) \times (\frac{2}{3} \text{ hour/car}) = 10.67 \text{ hours}$$

- With the new mechanic:

$$(16 \text{ cars/day}) \times (\frac{1}{4} \text{ hour/car}) = 4 \text{ hours}$$

- The total daily system costs are less with the new mechanic:

$$\$653.33 - \$360 = \$293.33 \text{ savings}$$

- Reducing waiting time is not the only way to reduce waiting cost
- Reducing the unit waiting cost (C_w) will also reduce total waiting cost
- This might be less expensive to achieve than reducing either W or W_q

Equations for the multichannel queuing model

- **Let**

$$m = \text{number of channels open}$$
$$\lambda = \text{average arrival rate}$$
$$\mu = \text{average service rate at each channel}$$

- Same basic assumptions as in the single-channel model

(12-13) $$P_0 = \frac{1}{\left[\sum_{n=0}^{m-1} \frac{1}{n!}\left(\frac{\lambda}{\mu}\right)^n\right] + \frac{1}{m!}\left(\frac{\lambda}{\mu}\right)^m \frac{m\mu}{m\mu - \lambda}} \quad \text{for } m\mu > \lambda$$

Probability that there are no people or units in the system.

(12-14) $$L = \frac{\lambda\mu(\lambda/\mu)^m}{(m-1)!(m\mu - \lambda)^2} P_0 + \frac{\lambda}{\mu}$$

Average number of people or units in the system.

(12-15) $$W = \frac{\mu(\lambda/\mu)^m}{(m-1)!(m\mu - \lambda)^2} P_0 + \frac{1}{\mu} = \frac{L}{\lambda}$$

Average time a unit spends in the waiting line or being serviced.

(12-16) $$L_q = L - \frac{\lambda}{\mu}$$

Average number of people or units in line waiting for service.

(12-17) $$W_q = W - \frac{1}{\mu} = \frac{L_q}{\lambda}$$

Average time a person or unit spends in the queue.

(12-18) $$\rho = \frac{\lambda}{m\mu}$$

Utilization rate.

- Arnold wants to investigate opening a second garage bay
- Hire a second worker at the same rate as the first worker
- Customer arrival rate remains the same: $\lambda = 2$, $\mu = 3$

$$P_0 = \frac{1}{\left[\sum_{n=0}^{1} \frac{1}{n!}\left(\frac{2}{3}\right)^n\right] + \frac{1}{2!}\left(\frac{2}{3}\right)^2\left(\frac{2\cdot 3}{2\cdot 3-2}\right)} = \frac{1}{1+\frac{2}{3}+\frac{1}{2}\left(\frac{4}{9}\right)\left(\frac{6}{4}\right)} = \frac{1}{2} = 0.5$$

Probability of 0 cars in the system

Arnold's Muffler Shop (M/M/m)

- Average number of cars in the system:

$$L = \frac{(2)(3)(2/3)^2}{(1)!(6-2)^2}\left(\frac{1}{2}\right) + \frac{2}{3} = \frac{8/3}{16}\left(\frac{1}{2}\right) + \frac{2}{3} = 0.75$$

- Average time in the system:

$$W = \frac{L}{\lambda} = \frac{0.75}{2} = 0.375 \text{ hours} = 22.5 \text{ minutes}$$

- Average number of cars in the queue:

$$L_q = L - \frac{\lambda}{\mu} = 0.75 - \frac{2}{3} = 0.083$$

- Average waiting time in the queue:

$$W_q = \frac{L_q}{\lambda} = \frac{0.083}{2} = 0.0417 \text{ hours} = 2.5 \text{ minutes}$$

Effect of Service Level on Operating Characteristics

Operating Characteristic	**One Mechanic** ($\mu = 3$)	**Two Mechanics** ($\mu = 3$)	**One Fast Mechanic** ($\mu = 4$)
P_0 (system empty)	0.33	0.50	0.50
L (cars in system)	2.00	0.75	1.00
W (time in system)	60 min	22.5 min	30 min
L_q (cars in queue)	1.33	0.083	0.50
W_q (time in queue)	40 min	2.5 min	15 min

- Adding the second service bay reduces waiting time but increases service cost
- Total daily waiting cost:

$$(8)(2)(0.0417)(50) = \$33.36$$

- Total daily service cost:

$$(8)(2)(15) = \$240$$

- Total daily system cost:

$$\$33.36 + \$240 = \$273.36$$

- Constant service times are used when customers are processed according to a fixed cycle
- Values for L_q, W_q, L, and W are always less than for variable service time models
- Both average queue length and average waiting time are halved in constant service rate models

Constant Service Time Model (M/D/1)

(12-19) Average length of the queue:

$$L_q = \frac{\lambda^2}{2\mu(\mu - \lambda)}$$

(12-20) Average waiting time in the queue:

$$W_q = \frac{\lambda}{2\mu(\mu - \lambda)}$$

(12-21) Average number of customers in the system:

$$L = L_q + \frac{\lambda}{\mu}$$

(12-22) Average time in the system:

$$W = W_q + \frac{1}{\mu}$$

M/D/1 example: Garcia-Golding Recycling, Inc.

- Company collects and compacts aluminum cans and glass bottles
- Trucks arrive at average rate of 8 per hour (Poisson distribution)
- Truck drivers wait about 15 minutes to unload
- Drivers and trucks cost \$60 per hour
- New automated machine processes truckloads at constant rate of 12 per hour
- New compactor amortized at \$3 per truck unloaded

Analysis of cost versus benefit

- Current waiting cost/trip: $\frac{1}{4}$ hour $\times$ \$60/hour = \$15/trip
- New system: $\lambda = 8$, $\mu = 12$
- Average waiting time: $W_q = \frac{8}{2\times 12\times(12-8)} = \frac{1}{12}$ hour
- Waiting cost/trip: $\frac{1}{12} \times 60 = \$5$/trip
- Savings: \$15 - \$5 = \$10/trip
- Equipment cost: \$3/trip
- Net savings: \$7/trip

- Models differ when population of potential customers is limited
- Dependent relationship between queue length and arrival rate
- Model assumptions:
  1. There is only one server
  2. The population of units seeking service is finite
  3. Arrivals follow Poisson distribution, service times exponential
  4. Customers served on FCFS basis

- Models differ when population of potential customers is limited
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  1. There is only one server
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  3. Arrivals follow Poisson distribution, service times exponential
  4. Customers served on FCFS basis

Equations for finite population model

- Let:

λ = mean arrival rate
μ = mean service rate
N = size of the population

(12-23) $$P_0 = \frac{1}{\sum_{n=0}^{N} \frac{N!}{(N-n)!}\left(\frac{\lambda}{\mu}\right)^n}$$

(12-24) $$L_q = N - \left(\frac{\lambda+\mu}{\lambda}\right)(1-P_0)$$

(12-25) $$L = L_q + (1-P_0)$$

(12-26) $$W_q = \frac{L_q}{(N-L)\lambda}$$

(12-27) $$W = W_q + \frac{1}{\mu}$$

(12-28) $$P_n = \frac{N!}{(N-n)!}\left(\frac{\lambda}{\mu}\right)^n P_0 \quad \text{for } n = 0, 1, \ldots, N$$

- Department has 5 printers needing repair after 20 hours of work
- Breakdowns follow Poisson distribution
- Technician services printers in 2 hours (exponential distribution)

$$\lambda = \frac{1}{20} = 0.05 \text{ printer/hour} \quad \mu = \frac{1}{2} = 0.50 \text{ printer/hour}$$

- Probability system is empty:

$$P_0 = \frac{1}{\sum_{n=0}^{5} \frac{5!}{(5-n)!} \left(\frac{0.05}{0.5}\right)^n} = 0.564$$

- Average queue length:

$$L_q = 5 - \left(\frac{0.05 + 0.5}{0.05}\right)(1 - 0.564) = 0.2 \text{ printer}$$

- Average waiting time:

$$W_q = \frac{0.2}{(5 - 0.64)(0.05)} = 0.91 \text{ hour}$$

- Average time in system:

$$W = 0.91 + \frac{1}{0.50} = 2.91 \text{ hours}$$

- Average printers down: $L = 0.2 + (1 - 0.564) = 0.64$
- Total hourly cost: $(0.64)(120) + 25 = \$101.80$

- Certain relationships exist for any queuing system in a **steady state**
- Steady state condition: System is in normal stabilized condition after initial transient state
- **Little's Flow Equations**:

$$L = \lambda W \quad (\text{or} \quad W = L/\lambda)$$
$$L_q = \lambda W_q \quad (\text{or} \quad W_q = L_q/\lambda)$$

- Third condition that must always be satisfied:

$$W = W_q + 1/\mu$$

Questions

